Practical 2. Foundations of the Brain

JXH-1104 Neuroscience of Sport Performance

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Get these slides:

Aim

Time a signal as it passes through a whole row of "nervous systems", and find out how much of a reaction is nerve conduction and how much is everything else.

Chain set up

Your row is a chain. Aim: 10 people. In each chain:

- The person at the beginning of the chain is the **starter**. The starter gives the first signal and does not react to anything.
- One person, the **timer**, stands outside of the chain where they can see both chain ends. The timer needs a stopwatch (e.g., through their phone).
- Everyone else in the chain **reacts**. That number is **N**. (Note: the starter and the timer are not counted in N.)

Signals (sound or sight) travel along the chain and is timed by the timer.

Predict

Everyone answers **question 1** in the form: how many milliseconds do you think each person will add to the chain? (Note: You have just seen a number in the lecture. Use it if you want to.)

Steps

Chains go one at a time, so the room stays quiet for whoever is running.

A. By sound

1. Starter and timer keep theirs open. Everyone who reacts closes their eyes.
2. The starter claps loudly once. The timer starts the stopwatch on that clap.
3. Each person claps as soon as they **hear** the person immediatey before them clap.
4. The timer stops the stopwatch on the **last** clap, and notes the time (in *seconds* e.g., 2.546).
5. Do this **three times**. The timer writes down all three.

B. By sight

6. Eyes open. Everyone turns to watch the person before them. Complete silence.
7. The starter raises a hand. Each person raises a hand as soon as they **see** the previous hand go up.
8. The timer stops on the **last** hand. Three trials, all three written down.

Submit

The **timer** submits on behalf of the whole chain: the number of members of the chain excluding the starter and the timer (i.e., **N**) and the six totals (3 values for each modality).

Open the form:

LINK

What to notice

Discuss in pairs first, then we look at the class data on the screen.

- Divide one of your totals by N, then multiply by 1000. That is milliseconds per person. How does it compare with the time from the lecture?
- Which was faster, sound or sight? Is the difference in the direction the lecture predicted?
- Did trial 3 come out faster than trial 1? What changed between them: the nervous system, or something else?
- The chain measures five stages of a reaction, N times over. Which stages does a clap add that a finger-on-a-key does not?

If you have time

- Work out how many milliseconds of your per-person time is nerve conduction, using the lecture's estimate (about 40 ms). What fraction is left over, and what is it?
- The timer's thumb reacted twice, once at the start and once at the end. Roughly how much error does that put into the total? And how much per person, after dividing by N?
- Sprinters are disqualified if they leave the blocks less than 100 ms after the gun. Given what you measured today, why is 100 ms the line?

If your chain's numbers went missing

Use this chain: N = 9. By sound: 2.91 s, 2.64 s, 2.48 s. By sight: 3.62 s, 3.40 s, 3.15 s.